

Scientific Formalization of Continuous Quantum Fluid Dynamics: The Agora Verification Protocols

SocrateAI Scientific Agora

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Note (2026-09-26). Several claims below have since been corrected or retracted; see `RETRACTIONS.md` in the repository for the full record. In summary: the “non-linear Quantum Volterra echo” (QVE-02) is not a physical echo (R2); the QV-01 paragraph below describing an RPA-driven “onset of Landau Damping” was never actually implemented that way and has since been replaced by an exact Padé root-finding method with a documented negative result in weak coupling (R6, R7); the Q-RIP-03 manifold is T^2 in the code, not $\mathbb{RP}^1$ as stated below, and $L_* = 4$ is a conjecture/definitional identity, not a proven protective bound (R3, R4); and the Q-RHK-02 “Theorem 22.6” attribution has been checked against the source and found mismatched (R8). The text below is kept as a historical record and is annotated in place rather than rewritten.

Abstract

This report documents the formalized mathematical physics protocols executed within the SocrateAI Scientific Agora. Strict adherence to the “Zéro Simulation Flottante” policy is maintained throughout: all continuous physical phenomena, including Landau Damping, non-linear Quantum Volterra echoes, and Roton scattering kernels, are extracted algebraically over the rational field $\mathbb{Q}$ avoiding any floating-point hallucinations. The resulting exact sequences are subsequently checked by the Lean 4 kernel utilizing the Mathlib `Rat` type, which certifies the exact rational arithmetic performed on those sequences (see the note above and `RETRACTIONS.md` for what this does and does not establish about the physics). This multi-stage pipeline orchestrates symbolic computation across specialized processing layers to achieve formal verification of that arithmetic.

1 Introduction

The primary objective of the SocrateAI Scientific Agora is the absolute formal verification of theoretical physics bounds. This removes the uncertainties surrounding conventional numerical solvers. We enforce this via a dual-stage workflow: the `LinearResponseStage` handles the exact quantum fluid extraction, while the `KineticStage` executes the theoretical constraint bounding and mapping to the exact rational field $\mathbb{Q}$ for Lean 4 theorem proving.

2 Protocol QV-01: Quantum Vlasov Zero-Sound Extraction

In the high-frequency limits of liquid ^3He , Zero-Sound waves propagate collisionlessly. The exact infinite rational series for the continuous base response (Lindhard function) is given by:

$$\chi_0(z) = \sum_{k=1}^{\infty} \frac{1}{2k+1} z^{2k} \tag{1}$$

where $z = \frac{qv_F}{\omega}$. Introducing the Landau interaction parameter F_0^s , the Random Phase Approximation (RPA) response is:

$$\chi_{RPA}(z) = \frac{\chi_0(z)}{1 - F_0^s \chi_0(z)} \quad (2)$$

Update (see RETRACTIONS.md R6/R7): the RPA-driven pole tracking described above as originally planned was never actually implemented – an early attempt fed a polynomial into a function expecting Taylor coefficients and produced meaningless output, and separately violated the exact-arithmetic rule via a float round-trip. It has been replaced by the following, which is implemented and tested. Instead of floating-point numerical integration, the exact diagonal $[M/M]$ Padé approximant of the Landau zero-sound kernel $\chi(s)$ is computed strictly over $\mathbb{Q}$, and the dispersion relation $\chi(s) = 1/F_0^s$ is solved exactly as a polynomial root. For strong coupling (e.g. $F_0^s = 93/10$, ${}^3\text{He}$ -like) this gives $s = \sqrt{37/10} \dots$ at $[1/1]$, agreeing with a 60-digit reference to 6.8×10^{-10} at $[4/4]$. For weak coupling, no admissible root exists at any order – the mode is exponentially close to the particle-hole continuum edge, $s - 1 \simeq 2e^{-2-2/F_0^s}$, and this negative result is characterised analytically rather than “tracked into” the continuum, which the method cannot do.

3 Protocol QVE-02: Second-order Volterra response (not an echo – see RETRACTIONS.md R2)

Demonstrating cross-domain Stage-to-Stage integration, the LinearResponseStage generates the $\mathcal{O}(\epsilon^1)$ linear density perturbation sequence $\rho^{(1)}(t)$. The KineticStage ingests this to compute the induced electric field $E^{(1)}(t)$, subsequently evaluating the exact Cauchy convolution product of the non-linear source term $S^{(2)} = \rho^{(1)} \cdot E^{(1)}$. This exact Volterra integral expansion yields the $\mathcal{O}(\epsilon^2)$ density sequence, which is the Taylor series of $\text{Si}(t)^2/2$ – not a plasma echo, which is a large-time phenomenon requiring two pulses at distinct wavenumbers, neither present here. The computation itself is exact and correct; only the “echo” label is retracted. This prevents discretization drift completely, securing the expansion for Lean 4.

4 Protocol Q-RHK-02: Roton Fractional Heat Kernels

To evaluate the stability of Pitaevskii Plateau phenomena in superfluid ${}^4\text{He}$, a phenomenological angular scattering kernel $\beta(\cos \theta)$ is extracted:

$$\beta(\cos \theta) = \frac{1}{2}(1 + \cos^2 \theta) \exp(-0.1(1 - \cos \theta)) \quad (3)$$

We compute the following combination, previously attributed to Villani’s Theorem 22.6 (arXiv:2501.00925) but now checked against the source and found mismatched — Theorem 22.6 is real but bounds a different, multiplicative quantity ($2\sqrt{d} \cdot \sqrt{m_r/M_r}$); the combination below is an independently-defined convention of this project (see RETRACTIONS.md R8):

- Spherical Curvature Integration: $\Sigma(\beta) \approx 1.899$
- Kernel Infimum and Supremum: $m_r \approx 0.4513$, $M_r = 1.0$
- Admissible Singularity Limit: $\gamma_{\text{bound}} \approx 1.9513$

These constants define the critical Fisher Information dissipation bound required to ensure smooth equilibration.

5 Protocol Q-RIP-03: 2D Quantum Ripplons (conjecture stage)

Evaluating topological phase-mixing of capillary waves (ripplons) on a 2D flat topology (T^2 in the implementation – an earlier draft of this document said $\mathbb{RP}^1$, a discrepancy recorded in RETRACTIONS.md R4 and corrected here), the differential Bakry-Émery curvature is conjectured, not derived: with $L_*(d) := 2d$ taken as a definition, $L_*(2) = 4$ follows by arithmetic alone. This is not a proven topological constant, and the claim that it “protects 2D quantum films from Landau damping collapse” has no support here – see RETRACTIONS.md R3 for the full caveat.

6 Conclusion

By leveraging an exact $\mathbb{Q}$ -based extraction layer and deploying subsequent proof resolution using Lean 4 `Mathlib` tactics (e.g. `norm_num`), the Agora eliminates numerical discrepancies. No simulated pseudo-data is introduced, strictly aligning with empirical rigor.